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STHKT: Spatiotemporal Knowledge Tracing with Topological Hawkes Process

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ABSTRACT

Knowledge Tracing (KT) is a method that seeks to forecast students' future performance based on their historical interactions with intelligent tutoring systems. Various KT techniques have been developed from distinct perspectives, such as probability, logic, and deep learning, to scrutinize interaction sequences. However, most current methods predominantly concentrate on temporal or structural information, considering simple question sequences or interaction sequences, but fail to describe the topological dependencies between sequences, which can lead to changes in causal structures, as referred to as dynamic structural information in the paper. The dynamic information includes the clustering effect among concepts and questions, as well as the group effect among students and questions: Questions with the same knowledge concepts tend to present increasingly similar response patterns in the sequence, and concurrently, students who answer similar questions often exhibit closer knowledge states. We introduce the STHKT method, which employs a tripartite heterogeneous graph to capture relationships among concepts, questions, and students. By incorporating temporal information through a topological graph, we employ two Graph Convolution Networks (GCNs) to model the two effects separately. Following this, we apply the Hawkes Process to introduce a spatiotemporal attention mechanism that combines both effects. Through comparative experiments, ablation studies and case studies, we validate the efficacy of key components in the model. Experiment results indicate that the STHKT model can effectively model the two major effects, integrate dynamic structural information, and to some extent improve the accuracy of predicting students' response probabilities. It is the first time that these two effects have been considered in the process of knowledge tracing, and it is adaptable to large-scale online education platforms.

1. Introduction

Knowledge Tracing, as defined in Abdelrahman et al. (2023), Liu et al. (2021), is a task aimed at modeling students' knowledge proficiency based on their learning interaction data. This is accomplished by observing students' responses to questions related to various concepts, thereby indirectly tracing their knowledge states on specific concepts. Generally speaking, most models simplify the learning environment by presuming that students' responses are binary (*i.e.*, correct or incorrect). Given a new question, the output of the knowledge tracing model is the probability of students answering it correctly. This is one of the most critical components in constructing adaptive education systems. In adaptive education systems, whether it is precise recommendation,

students' learning path planning or knowledge graph construction, the first step is to accurately predict a students' knowledge proficiency level (Cui & Sachan, 2023; Luaces et al., 2017).

The field of knowledge tracing has seen continuous development through the introduction of innovative methodologies. Bayesian Knowledge Tracing (BKT) (Yudelson et al., 2013) was the first method proposed for mining students' knowledge states, utilizing the Hidden Markov Model. With the maturation of deep networks, Deep Knowledge Tracing (DKT) (Piech et al., 2015) was introduced, simulating students' learning process using RNN/LSTM (Lipton et al., 2015). Since then, various neural network methods have been employed for knowledge

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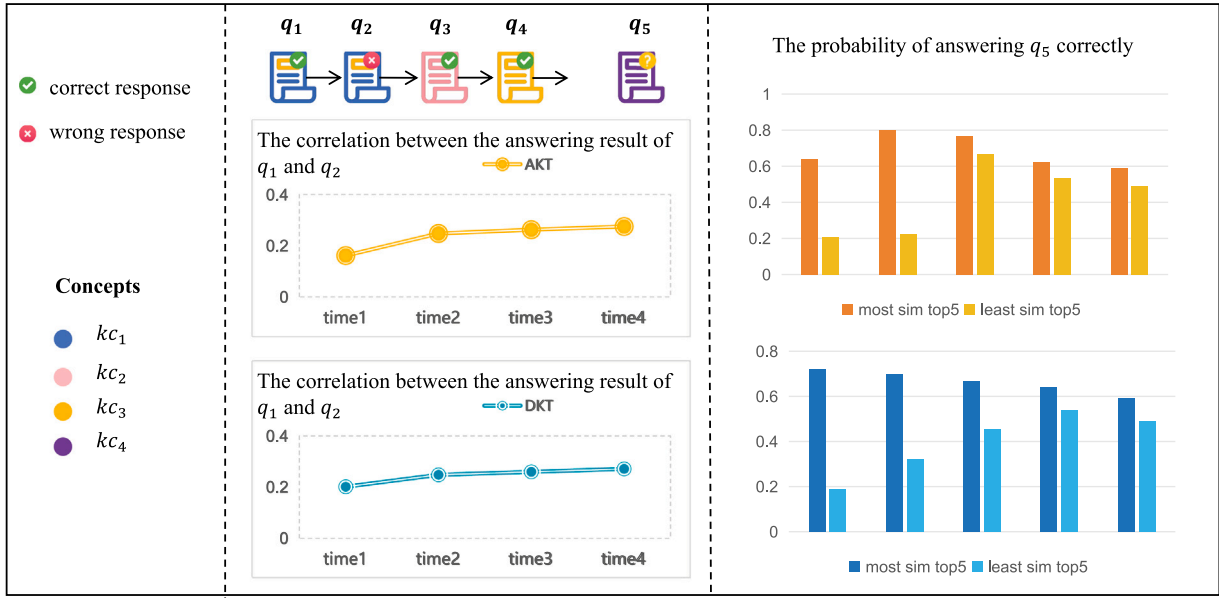


Fig. 1. A toy example of the process of knowledge tracing which traced by AKT and DKT, which in the visualization data on the right, orange represents AKT, while blue represents DKT. In the middle of the figure, we use $p(q_1, q_2) \log_2 \frac{p(q_1, q_2)}{p(q_1)p(q_2)}$ to calculate the correlation score of q_1 and q_2 , and we can find the two questions which have the same concepts, are increasingly related. On the right side of the figure, we choose the five most similar sequences and the five least similar sequences. Then we predict the correct probability of q_5 , and find the five sequences which are the most similar have the more similar prediction.

tracing. One particularly powerful branch is the attention-based knowledge tracing model (Ghosh et al., 2020). Due to the inherent matching of dependency relationships between knowledge concepts, researchers have employed Graph Neural Networks (GNN) to utilize the graph structure within knowledge concepts for knowledge tracing (Nakagawa et al., 2019). In practical terms, these advancements significantly impact educational technology. Models using attention mechanisms and graph neural networks improve the tracing of students' understanding, offering precise feedback and tailored recommendations. Integrating these methods into tutoring systems allows real-time support and more personalized instruction, enhancing overall student outcomes and contributing to individualized learning.

However, despite the notable success of the aforementioned methods, these approaches always focus on temporal or structural information, considering simple question sequences or interaction sequences. They often fail to adequately describe the topological dependencies between these sequences, which can lead to changes in causal structures, namely dynamic structural information in the paper. This is evident in two distinct effects: (1) **the knowledge concept clustering effect: questions sharing the same knowledge concepts tend to present increasingly similar response patterns in the sequence.** As illustrated in Fig. 1, we quantify the correlation between q_1 and q_2 across various different timestamps by calculating the similarity of the probability of correct answer on them between q_1 and q_2 at different moments using Mutual Information: $p(q_1, q_2) \log_2 \frac{p(q_1, q_2)}{p(q_1)p(q_2)}$ (Shannon, 1948). If q_1 and q_2 are completely independent, their joint probability $p(q_1, q_2)$ equals the product of their individual probabilities $p(q_1)$ and $p(q_2)$. At this point, the correlation is zero, indicating that there is no information sharing between q_1 and q_2 . If q_1 and q_2 are completely related, then $p(q_1, q_2) = p(q_1) = p(q_2)$, at this time the correlation is at its maximum, indicating that the information between q_1 and q_2 is completely shared. Therefore, this formula can effectively measure the correlation between two questions. As students continue to solve questions, question q_1 and q_2 that share the same knowledge concepts show a gradually increasing similarity in their responses. (2) **the student group effect: students who answer similar questions often demonstrate similar knowledge states, i.e., the higher the overlap of answered questions, the closer the students' knowledge states tend to be.** In Fig. 1, we select the five sequences that are most

similar to the sequence shown in the middle and the five least similar sequences, and respectively evaluate the correctness rate for q_5 . We observe that the probability of correctly answering q_5 is relatively close among the five most similar sequences, and they are all generally high. This implies that the sequence exemplified in the figure, along with the five most similar sequences, possess relatively similar knowledge states.

We assert that it is vital and advantageous to contemplate the concepts of clustering effect and student group effect within the context of knowledge tracing. In this paper, we have proposed a Spatiotemporal Knowledge Tracing framework (STHKT¹) based on a Topological Hawkes Process to better model the two effects mentioned above. Firstly, to obtain the structural information in the knowledge tracing process, we construct a tripartite heterogeneous graph comprising of Concepts, Questions and Students (Su et al., 2022). Secondly, to combine structural information with temporal information, we topologize the question sequences and associated concepts and students within the tripartite heterogeneous graph, thereby creating topological graphs. During the topologization of a question sequence, we select question nodes from the tripartite heterogeneous graph and generate directed edges according to temporal relationships. Then, we identify all adjacent concept and student nodes based on these question nodes. If there is a common concept or student node between two question nodes, the earlier one points to the shared node, followed by the later question node, thereby forming a topological graph. Thirdly, to model the influence of clustering effect and group effect with advantage, we employ a graph convolutional method. This approach fundamentally captures cumulative effects (Cai et al., 2022; Defferrard et al., 2016; Shang & Sun, 2019). Ultimately, given that Hawkes processes and the convolution process of topological graphs are essentially identical, we combine the results of modeling the two effects with a monotonic attention mechanism, using Hawkes process. It enhances the monotonic attention to a spatiotemporal attention mechanism. By leveraging the characteristics of topological graphs and Hawkes process, we integrate temporal information into the knowledge tracing process, combining it with the two effects to achieve higher prediction accuracy.

Our main contributions are summarized as follows:

¹ <https://github.com/Ting1024/STHKT3.git>

- Construct a tripartite heterogeneous graph that incorporates concepts, questions, and students. Thoroughly analyze the clustering effect and the group effect.
- Topologize the question sequences, related concepts, and students in the tripartite heterogeneous graph to generate topological graphs. Employ two graph convolution networks to model clustering effect and group effect separately.
- Optimize the attention mechanism using the intensity function of the topological Hawkes process. This optimization enables the fusion of information from both effects into the knowledge tracing process.

2. Related works

2.1. Knowledge tracing

Knowledge Tracing is a technique that models and predicts a student's mastery level, typically by analyzing their exercise accuracy. The primary goal of Knowledge Tracing is to provide more precise predictions for each student, enabling the allocation of tailored learning resources based on individual needs. Knowledge Tracing is a crucial technology for the development of adaptive education systems.

As deep learning methods continue to evolve, they are increasingly utilized in the field of knowledge tracing (Huang et al., 2020). There are two different types of knowledge tracing methods, each focusing on different aspects: Sequence-Centric KT and Graph-Enhanced-Centric KT. In the meanwhile, the sparsity is also an important issue to be studied for not only Sequence-Centric KT, but also Graph-Enhanced-Centric KT within education field.

2.1.1. Sequence-Centric KT

In the field of Sequence-Centric Knowledge Transfer, Attentional Knowledge Tracing (AKT) (Ghosh et al., 2020) is one of the most significant methodologies. The attention mechanism has rapidly gained prominence due to its three key advantages: fewer parameters, faster processing speed, and superior performance. AKT (Ghosh et al., 2020) represents the question sequence as $q_0, q_1, \dots, q_t$, while the interaction sequence is represented as $\{(q_0, a_0), (q_1, a_1), \dots, (q_{t-1}, a_{t-1})\}$. Following the application of a monotonic attention mechanism, we can obtain personalized understanding information $\hat{x}_t$ based on the question sequence. Similarly, we can obtain personalized ability information, denoted as $\hat{y}_t$, based on the interaction sequence.

Recently, numerous cutting-edge methods have been proposed. For example, (Yin et al., 2023) introduced contrastive learning techniques to reveal examples of semantic similarity or dissimilarity in learning history, promoting a better understanding of the relationship between students and questions. Wang et al. (2023) utilized two novel attention-based models to directly disambiguate the relationships between each item in the input sequence, addressing the prediction instability observed in Deep Knowledge Tracing (DKT) (Piech et al., 2015) by state automaton (FSA).

2.1.2. Graph-Enhanced-Centric KT

Recent research has shown that graph structures are highly effective for knowledge tracing, leading to the development of Graph-Enhanced Knowledge Tracing (GKT). The natural graph relationships among concepts allow for the integration of relational inductive biases from knowledge graphs, which improves the performance of knowledge tracing systems. A method known as Graph Knowledge Tracing (GKT) (Nakagawa et al., 2019) has been proposed, which models the latent graph structure of concepts as $G = (V, E)$, with V representing the set of concepts, $E \subset V \times V$ the relationships between them, and $h_t = \{h_{i \in V}^t\}$ the students' temporary knowledge state at time t . The method updates the students' knowledge state by aggregating the state and embeddings of both answered and adjacent concepts as they

progress through questions, using this information to predict future performance.

Zhang et al. (2023) optimized Deep Knowledge Tracing (DKT) (Piech et al., 2015) by incorporating a hierarchical concept tree to integrate the description of the concept structure in problems; (Jiang et al., 2023) addressed the issue of Graph Neural Networks (GNN) failing to capture complex higher-order and heterogeneous relationships between questions and concepts. They employed Metapath2vec (Dong et al., 2017) to embed a heterogeneous graph of question concepts and designed two attention-based encoders to represent learners' engagement and knowledge states.

To address the neglect of complex relationships between exercises, the introduction of a dual graph structure and ensemble learning method aims to comprehensively capture the associations and interaction transitions between exercises during students' learning processes, thereby enhancing the accuracy and effectiveness of knowledge tracing (Cui et al., 2024). Due to difficulties in feature extraction caused by the complexity of knowledge structures and the sparsity of data in existing KT models, which limits their effectiveness and interpretability, (Ni et al., 2023) proposes a new method called HHSKT that utilizes hierarchical heterogeneous knowledge structures and attention mechanisms. With students only being exposed to a small number of questions from a vast question bank, (Wu & Ling, 2023) employs a heterogeneous hypergraph network (HHN) (Khan et al., 2023) and introduces an attention mechanism to integrate information, enhancing question representation through self-supervised learning and improving the model's ability to capture latent relationships between questions.

2.1.3. The sparsity problems of knowledge tracing

In knowledge tracing, data sparsity is a significant issue as students often interact with only a small subset of the question pool or as a question may involve only one or a few concepts. This sparsity not only affects the performance of knowledge tracing models but also makes it challenging for models to extract meaningful information from limited data. To address these challenges, several strategies have been developed. For example, data augmentation and synthetic data generation methods (Wen et al., 2020) aim to enrich sparse datasets by creating additional simulated interactions. Contrastive learning methods (Yin et al., 2023) have been used to improve the model's ability to discern meaningful patterns from limited data by emphasizing semantic similarity and dissimilarity. Recent advances also include the use of advanced attention mechanisms and graph-based models (Wang et al., 2023), which better capture relationships within sparse datasets. Despite these innovations, effectively managing data sparsity remains a crucial challenge in knowledge tracing, highlighting the need for ongoing research and development in this area.

2.2. Hawkes process

The Hawkes Process (Hawkes, 1971), introduced by Alan G. Hawkes in 1971, is a mathematical model utilized to describe the timing of events. Specifically, it is employed within the field of point processes. This self-exciting process indicates that the intensity of event occurrences is influenced by the past events.

In a Hawkes Process, each event has an impact on the rate of future occurrences (Luo et al., 2015). The intensity function of the process captures this influence and can vary over time. This intensity function is typically modeled as a sum of baseline intensity and triggering functions (Zhou et al., 2013), representing the influence of the past events on the future occurrences.

The intensity function in the original Hawkes Process paper is represented by the following formula:

$$\lambda(t) = \lambda_0(t) + \sum_{t_i < t} \alpha(t - t_i). \quad (1)$$

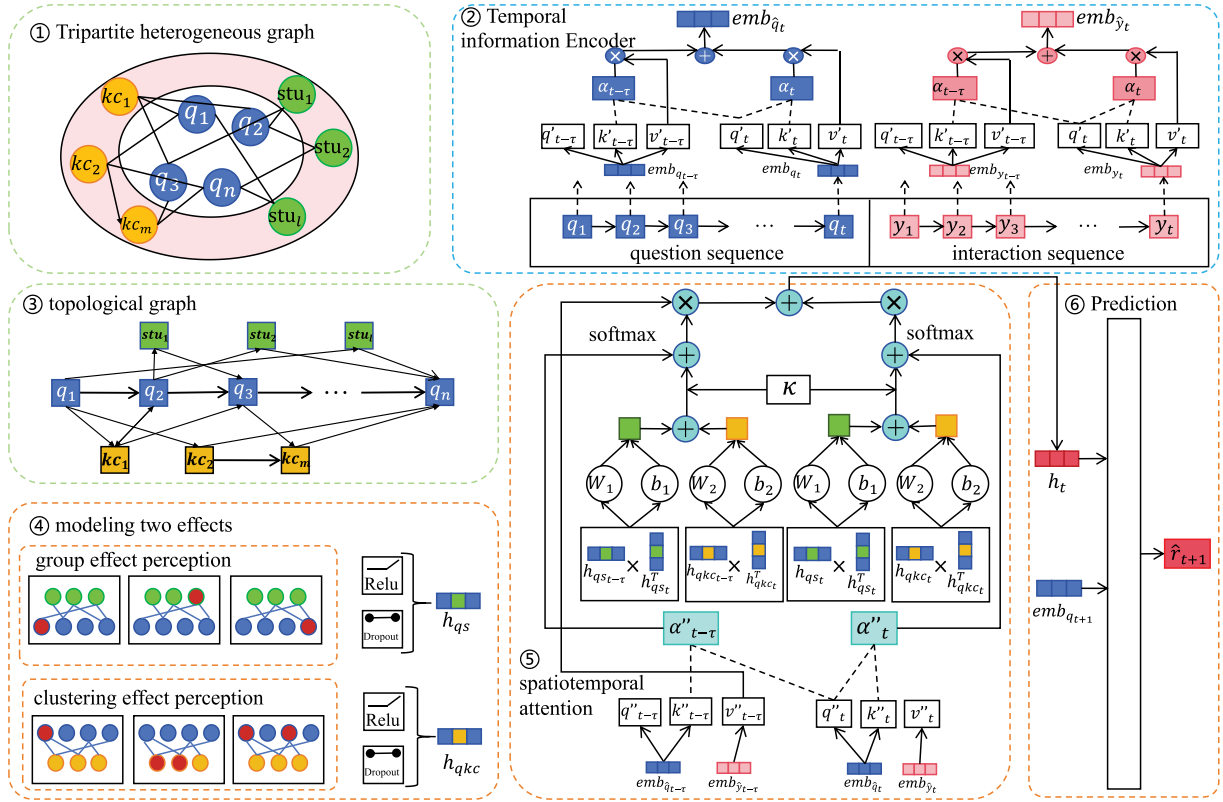


Fig. 2. The Architecture of STHKT. We begin by constructing a tripartite heterogeneous graph from the dataset, which establishes relationships among concepts, questions, and students. During the learning process, we use a monotonic attention mechanism to encode the temporal information within the question and interaction sequences. Utilizing the tripartite heterogeneous graph and the temporal information, we topologize each question sequence, resulting in a topological graph for each sequence. Subsequently, we utilize two Graph Convolutional Networks (GCNs) to model two effects respectively in the topological graph. There are two convolution kernels involved, one for the question-concept subgraph and the other for the question-student subgraph, modeling the two effects respectively. Employing the outcomes from modeling the two effects, we enhance the monotonic attention mechanism to a spatiotemporal attention mechanism via the Hawkes process. Finally, we use a prediction module to predict the answer results based on the processed information.

$\lambda(t)$ represents the intensity of events at time t . $\lambda_0(t)$ represents the baseline intensity, which represents the expected rate of events occurring at time t independently of past events. t_i represents the previous event times that are smaller than the current time t . $\alpha(t - t_i)$ is the triggering function, which characterizes the influence of past events on the occurrence rate of future events. It measures the impact of each past event on the intensity of events at time t . The intensity function $\lambda(t)$ integrates the baseline intensity $\lambda_0(t)$ and the cumulative effect of triggering functions $\alpha(t - t_i)$ for past events to determine the overall intensity at time t .

The Hawkes Process has been widely utilized across various domains, including knowledge tracing. In this context, Hawkes process has been utilized to capture the sequential nature of students' learning activities (Wang et al., 2021). By modeling the temporal dependencies and interactions between different learning events, such as answering questions or watching instructional videos, Hawkes process can provide a more accurate estimation of students' knowledge states and their learning progress over time. The triggering functions capture the impact of past events on the occurrence rate of future events, enabling a dynamic and personalized modeling of students' learning behaviors.

In the context of the Hawkes process, a topological graph can be employed to model the interaction between events (Cai et al., 2022). Each event is represented as a node in the graph, with edges signifying the influence or triggering relationship between events. The intensity of each event is determined by the combination of the baseline intensity and the cumulative effect of triggering functions associated with past events connected through the graph. This approach leverages the graph structure to provide additional information about the dependencies and interactions among events, thereby enhancing the modeling capabilities of the Hawkes process and get a better performance.

3. Method

3.1. Problem definition

In a Knowledge Tracing task, suppose there are $|S|$ students, $|Q|$ questions, and $|K|$ concepts where students do questions individually. We record a student response as $s = \{(q_1, a_1), (q_2, a_2), \dots, (q_t, a_t)\}$, and $q_t \in Q$ is represented as the student finished the question q_t in the timestamp t . $a_t \in \{0, 1\}$ is represented as the student answer about question q_t and t is the total step, where $a_t = 1$ if the student answers q_t correctly, and $a_t = 0$ otherwise. In addition, we also need concepts related to the question, so we need to expand s to $s^* = \{(q_1, kclist(q_1), a_1), (q_2, kclist(q_2), a_2), \dots, (q_t, kclist(q_t), a_t)\}$, where $kclist(q_t) \subseteq K$.

The task is to predict the correct probability of next question q_{t+1} according previous response sequence. And it can be formulated as $p(\hat{a}_{t+1} = a_{t+1} | q_{t+1}, s^*)$.

3.2. Architecture

As depicted in Fig. 2, we initially construct a tripartite heterogeneous graph composed of concept, question, and student nodes to establish structural information. During the learning or prediction process, we employ a monotonic attention mechanism to process the question sequence and interaction sequence to obtain temporal information. To integrate temporal information with structural information within the tripartite heterogeneous graph, we employ a topological graph of the questions within the sequence, with correlative concepts

and students serving as neighboring nodes based on the tripartite heterogeneous graph. Then, we utilize graph convolution method (Zhang et al., 2019) to model the clustering and group effects. One convolution kernel is used in the question–concept subgraph, while the other in the question–student subgraph. After modeling the two effects, we employ the intensity function of the Hawkes process to integrate the monotonic attention mechanism with the result of modeling the two effects, which constitutes spatiotemporal attention mechanism. It allows us to calculate an attention score not only based on the interaction sequence, but also on the aforementioned effects. Ultimately, our model utilizes the state embedding h_t and the next question embedding $emb_{q_{t+1}}$ to predict the probability of q_{t+1} being correct.

Based on the order in which each module is executed, in the following subsection, we will discuss the construction of a tripartite heterogeneous graph, temporal information encoding, topology graph construction, modeling of the two major effects, and spatiotemporal information encoding.

3.2.1. Tripartite heterogeneous graph

In this study, the purpose of this module is to establish the structural relationships among students, questions, and concepts. We initially construct a tripartite graph using the relationships of concept–question, concept–concept and question–student, which constructs the structural information. The process which is illustrated in the top left corner of Figs. 2 and 3.

Suppose that K , Q , and S represent the groups of Concepts, Questions, and Students in the network. $K_j \in K$ represents a concept, $j = 1, 2, 3, \dots, n_k$; $Q_i \in Q$ represents a question, $i = 1, 2, 3, \dots, n_q$; $S_m \in S$ represents a student, $m = 1, 2, 3, \dots, n_s$. The dataset contains a variety of interaction records, typically presented as interaction sequences. As shown in Fig. 3, in the case of student stu_i , a sequence of questions $\{q_1, q_6, \dots, q_n\}$ and the corresponding concept sequence $\{kc_1, kc_2, \dots, kc_m\}$ can be found. According to sequence reader, we can extract pairs from concepts and questions, as well as questions and students. In some datasets where dependencies among concepts exist, pairs of concepts, such as $\langle kc_2, kc_m \rangle$ in Fig. 3, can be generated through the Dependency Reader.

Using the generated heterogeneous pairs, we can construct a tripartite heterogeneous graph. Simplifying this graph to an unweighted and undirected format, an edge exists between a question q_i and a concept kc_j if the pair $\langle q_i, kc_j \rangle$ is present. Similarly, an edge exists between a student s_m and a question q_l if the pair $\langle q_l, s_m \rangle$ exists. To represent the dependency relationship between each pair of concepts, the heterogeneous graph assigns a value of $E_{ij} = 1$ if concept kc_i depends on concept kc_j . This indicates that the pair $\langle kc_i, kc_j \rangle$ exists in the graph.

3.2.2. Temporal information encoder

The purpose of the module is to model the temporal information through interaction sequences.

Since the subsequent construct a topological graph will require temporal information, we utilize two Transformer encoders only with monotonic attention mechanism to process the question sequence $\langle q_1, q_2, \dots, q_t \rangle$ and interaction sequence $\langle (q_1, a_1), (q_2, a_2), \dots, (q_t, a_t) \rangle$ separately. Following the processing by these temporal information encoders, we acquire the embeddings that represent the understanding of historical information for each moment, such as $emb_{\hat{q}_t}$, and the mastery information of the question such as $emb_{\hat{q}_t}$.

3.2.3. Topological graph

The purpose of this module is to obtain corresponding topological graphs based on different sequences for subsequent effect modeling.

After getting structural information such as Tripartite Heterogeneous Graph and temporal information such as the result of encoding a sequence, we can utilize a topological graph to combine the two types of information.

While topologizing a sequence of questions, we identify the question nodes within the tripartite heterogeneous graph and establish directed edges reflecting temporal relationships within the sequence. Subsequently, we extend the selection to include all concept nodes and student nodes adjacent to the chosen question nodes in the tripartite heterogeneous graph. If there exists a common concept node or student node between two question nodes, the question node occurring earlier in the sequence is linked to the shared concept or student node, followed by a connection to the subsequent question node. This iterative process leads to the formation of a topological graph. As shown in the topological graph part in Fig. 2, a topological graph is obtained after topological transformation.

3.2.4. Modeling two effects with graph convolution kernel

The purpose of this module is to model two effects through the application of graph convolution methods to the topological graphs.

In the traditional Hawkes process, we formulate its Conditional Intensity Function as:

$$\lambda(t) = \lambda_0(t) + \sum_{t_i < t} \alpha(t - t_i). \quad (2)$$

We consider that every response to a question constitutes an event, with $\lambda(t)$ serving as the measure of event intensity at any given time t . $\lambda_0(t)$ is the baseline intensity, signifying the anticipated rate of events occurring at time t without regard for the past events. t_i refers to the timestamps of prior events that precede the current time t . The triggering function, $\alpha(t - t_i)$ characterizes the influence of past events on the occurrence rate of future events. It measures the impact of each past event on the intensity of events at time t .

And then, we use the properties of the Poisson distribution, that is, within a small time interval Δt , the probability of an event occurring is approximately equivalent to $\lambda(t)\Delta t$. Subsequently, we can divide the time axis into many small time intervals, and examine the probability of an event occurring within each time interval. Supposing having observed n events, which occurred at $t_1, t_2, \dots, t_n$, the probability of an event occurring in the subsequent time interval $[t_n, t_n + \Delta t]$ can be expressed as:

$$\begin{aligned} P(N(t_n + \Delta t) - N(t_n)) &= 1/(H(t_n)) \\ &= \lambda(t_n)\Delta t + o(\Delta t), \end{aligned} \quad (3)$$

where $H(t_n)$ represents the historical record of events that occurred at time t_n , while $N(t)$ denotes the total number of events that have occurred up to time t , and $o(\Delta t)$ is an infinitesimal of Δt .

Then, combined formula (2) with formula (3), we can get the formula (4) and (5):

$$\lambda(t_n) = \lambda_0(t_n) + \sum_{t_i < t_n} \alpha(t_n - t_i). \quad (4)$$

$$\begin{aligned} P(N(t_n + \Delta t) - N(t_n)) \\ &= (\lambda_0(t_n) + \sum_{t_i < t_n} \alpha(t_n - t_i))\Delta t + o(\Delta t). \end{aligned} \quad (5)$$

In the formula (5), the probability of an event occurring at time t_n in a very short time interval Δt , namely $P(N(t_n + \Delta t) - N(t_n))$. This probability is influenced by both the fundamental probability of the event itself and the accumulation of historical influences.

In the same time, Graph Convolution Network can be formulated as follow:

$$X^{(l+1)} = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} X^{(l)} W^{(l)}), \quad (6)$$

where $X^{(l+1)}$ is the node feature matrix of the $l + 1$ -th layer. $\tilde{A}$ denotes the adjacency matrix augmented with self-loops, and $\tilde{D}$ is the degree matrix of $\tilde{A}$. $W^{(l)}$ is the weight matrix of the l th layer, and σ is the activation function.

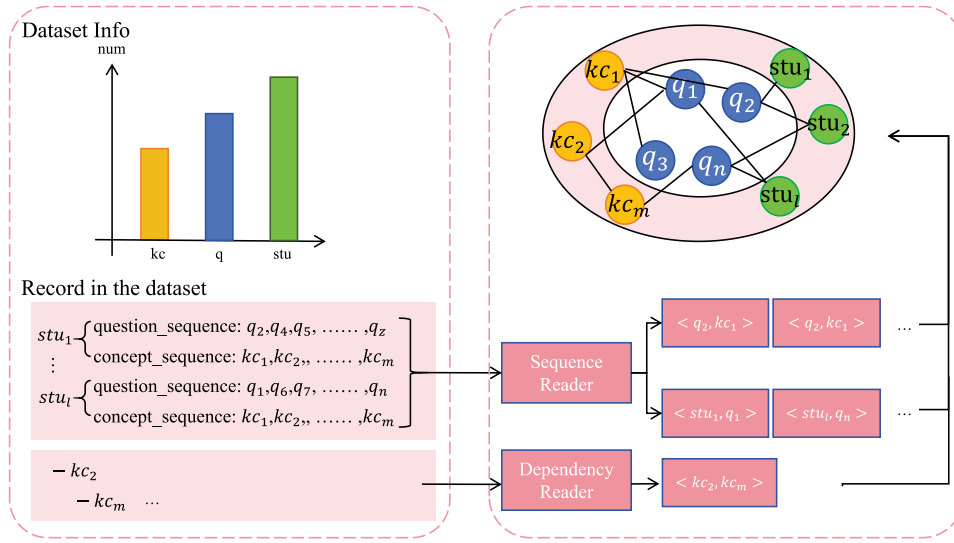


Fig. 3. The construction of Tripartite Heterogeneous Graph. A dataset contains a variety of interaction records, typically presented as interaction sequences. For instance, in the case of student stu_i , a sequence of questions $\{q_1, q_2, \dots, q_n\}$ and the corresponding concept sequence $\{kc_1, kc_2, \dots, kc_m\}$ can be found. The Sequence Reader can be used to form pairs of concepts and questions, as well as questions and students. In datasets where there are dependencies among concepts, pairs of concepts can be generated through the Dependency Reader. These pairs are utilized to construct a tripartite heterogeneous graph.

Next, we expand the above formula (6) into the aggregation process for each node.

$$\begin{aligned}
 x_t^{(l+1)} &= \sigma \left(\sum_{i \in \mathcal{N}(t) \cup t} \frac{1}{\sqrt{\tilde{d}_t \tilde{d}_i}} x_i^{(l)} w^{(l)} \right) \\
 &= \sigma \left(\frac{1}{\sqrt{\tilde{d}_t}} x_t^{(l)} w^{(l)} + \frac{1}{\sqrt{\tilde{d}_t}} \sum_{i \in \mathcal{N}(t)} \frac{1}{\sqrt{\tilde{d}_i}} x_i^{(l)} w^{(l)} \right) \\
 &= \sigma \left(\frac{1}{\sqrt{\tilde{d}_t}} x_t^{(l)} w^{(l)} + \frac{1}{\sqrt{\tilde{d}_t}} \bar{x}_0^{(l)} w^{(l)} \right) \\
 &= \sigma \left(\frac{1}{\sqrt{\tilde{d}_t}} (x_t^{(l)} - \bar{x}_0^{(l)}) w^{(l)} \right), \quad (7)
 \end{aligned}$$

where $x_t^{(l+1)}$ is the feature vector of node t in the $(l+1)$ -th layer. $\mathcal{N}(t)$ is the set of neighbor nodes of node t , $\tilde{d}_t$ is the degree of node t and $x_i^{(l)}$ is the feature vector of node i in the l -th layer. $w^{(l)}$ is the weight vector of the l -th layer. $\bar{x}_0^{(l)}$ is the average feature vector of the neighboring nodes of node t in the l -th layer, which can be formulated as follows:

$$\bar{x}_0^{(l)} = \frac{1}{|\mathcal{N}(t)|} \sum_{i \in \mathcal{N}(t)} \sqrt{\tilde{d}_i} x_i^{(l)}.$$

From the formulas referenced earlier in (5) and (7), it is evident that the intensity function of Hawkes process and graph convolution operations (Song et al., 2021) fundamentally aggregate and update information for specific nodes based on certain conditions. This suggests that the intensity function can be integrated with the graph structure to simultaneously incorporate structural information into the intensity function.

3.2.5. Spatiotemporal attention mechanism

The purpose of this module is to combine the previously obtained temporal and spatial information (information after modeling the two effects) and extend the attention mechanism to obtain spatiotemporal attention mechanisms.

As above, our model mainly relies on Transformer Encoder to process the sequence, where the attention score calculation formula is usually designed as:

$$\text{Attention}(Q', K', V') = \text{softmax} \left(\frac{Q' K'^T}{\sqrt{d'_k}} \right) V'. \quad (8)$$

However, traditional attention mechanisms only take into account the personalized differences introduced by sequence changes, disregarding the clustering effect of concepts–questions and the group effect

of questions–students in historical interactions. Furthermore, building on the introduction of the adaptability of Hawkes process in Knowledge Tracing (Wang et al., 2021) at the beginning of this chapter, we employ the intensity function method to enhance the calculation of attention.

$$\lambda(q_{j,i}) = \text{softmax}(\lambda_0^{q_j, q_i} + \alpha_{q_j, q_i} \cdot \kappa_{q_j, q_i}(t_i - t_j)), \quad (9)$$

$$\lambda_0^{q_j, q_i} = \frac{Q' K'^T}{\sqrt{d'_k}} [j, i], Q' = K'. \quad (10)$$

As illustrated above, formula (9) represents the attention weight $\lambda(q_{j,i})$ associated with question event $j < i$ occurring at time points j and i . This weight is obtained by combining the basic attention weight $\lambda_0^{q_j, q_i}$ and the Hawkes Process intensity change function. formula (10) demonstrates the calculation method of the basic attention score. Given the monotonic nature of the attention score, we utilize $Q' = K'$.

$$h_{q_{kc}}[q_j, q_i] = \sigma(\tilde{D}_1^{-1/2} \tilde{A}_1 \tilde{D}_1^{-1/2} q W_1)[j, i], \quad (11)$$

$$h_{q_{qs}}[q_j, q_i] = \sigma(\tilde{D}_2^{-1/2} \tilde{A}_2 \tilde{D}_2^{-1/2} q W_2)[j, i]. \quad (12)$$

The formula (11) and (12) illustrate the process of obtaining the feature representations after dynamic structure convolution in the two bipartite heterogeneous graphs. As stated in the equation (6), $\tilde{A}_1$ and $\tilde{A}_2$ is the adjacency matrix augmented with self-loops, while $\tilde{D}_1$ and $\tilde{D}_2$ is the degree matrix of $\tilde{A}_1$ and $\tilde{A}_2$ respectively. q and $[j, i]$ represent the features of questions q_j and q_i and W_1 and W_2 represent two weight matrices respectively.

$$\begin{aligned}
 \alpha_{q_j, q_i} &= (W_{a_1} \cdot (h_{q_{kc}} \times h_{q_{kc}}^T) + b_{a_1} + \\
 &W_{a_2} \cdot (h_{q_{qs}} \times h_{q_{qs}}^T) + b_{a_2}) [q_j, q_i]. \quad (13)
 \end{aligned}$$

In formula (13), α_{q_j, q_i} represents the degree to which q_j influences the response in the target interaction q_i . The time point i and time point j are non-interchangeable, as there could be a risk of leakage.

$$\kappa_{q_j, q_i} = \exp(-(1 + \beta_{q_j, q_i}) \log(t_i - t_j)). \quad (14)$$

The formula (14) represents the decay component of the intensity function κ . The parameter β is a learnable variable that controls the decay behavior, which is designed based on the discrete time distance.

$$\beta_{q_j, q_i} = W_\beta \cdot \log(t_i - t_j). \quad (15)$$

The enhanced results suggest that by integrating topological Hawkes process, the dynamic fusion of tripartite heterogeneous graph structural information into the Transformer-based processing of interaction sequences facilitates knowledge tracing that captures dynamic structural information.

3.3. Loss function

The primary objective of this model is to forecast the learner's responses to subsequent questions. In the final prediction module, the input is the student's knowledge state and the embedding of the next question, denoted as q_{t+1} . This data is then processed through a fully connected layer and a sigmoid function to generate the probability $\hat{r}_{t+1} \in [0, 1]$ of answering the question correctly. Ultimately, the model parameters are trained using binary cross-entropy loss.

$$\mathcal{L} = -\frac{1}{N} \sum_{i=0}^N [y_{t+1} \log(\hat{r}_{t+1}) + (1 - y_{t+1}) \log(1 - \hat{r}_{t+1})], \quad (16)$$

where y_{t+1} represents the label of the question q_{t+1} in time $t + 1$ and N is the total number of questions that are learned.

3.4. Analysis of computational complexity

Firstly, in different sequence processes, we would topologize the tripartite heterogeneous graph composed of concepts, questions, and student nodes. That is to say, the entire topology graph only contains the question nodes in the student's problem-solving sequence, the concept nodes contained in the question nodes, and the student nodes. Since the dataset for knowledge tracing is often sparse, and the length of the problem-solving sequence is limited and controllable, the total number of nodes in the topology graph can be denoted as a not large variable $|V|$.

Moreover, the Hawkes process is a self-exciting process used to model the event dependence between events. In knowledge tracing, this is reflected in the mutual dependence between student interaction sequences. In the worst case, the time complexity and space complexity of the Hawkes process can reach $O(|V|^2)$ and $O(|V|)$, where $|V|$ represents the total number of nodes in a topology graph corresponding to an interaction sequence.

In addition, the time complexity of GCN usually depends on the sparsity of the graph. For sparse graphs, the time complexity and space complexity of graph convolution are usually expressed as $O(|E|)$ and $O(|V| + |E|)$, where $|V|$ and $|E|$ are the number of nodes and edges in the topology graph respectively. And our GCN network is applied on the topology graph, the number of its nodes and edges are controllable variables.

Although combining GCN with the Topological Hawkes Process increases computational complexity, the number of nodes and edges in the topology graph is not large and manageable, thus the impact on computational complexity is minimal. During experimentation, it has been observed that adding the Topological Hawkes Process and GCN does indeed increase convergence time and computational space. However, these effects are manageable and feasible.

4. Experiments

4.1. Experimental setup

4.1.1. Dataset benchmark

The experimental datasets come from the real world, where we select **three** classic datasets: Assistment2009, Assistment2017 and Aaai23 (see Table 1). :

- **Assistment2009**²: Assistment2009 is the ASSISTment data collected in the 2009–2010 academic year. It is split into two distinct files: one comprises all skill builder data, and the other includes all non-skill builder data. We elected to utilize the skill builder data file. This dataset is a sparse dataset, with a question–concept density of about 0.97% and question–student density of about 0.38%.
- **Assistment2017**³: This data is obtained from the ASSISTment 2017 competition. It constitutes the richest dataset in terms of interaction count, featuring 942,816 interactions, 686 students, and 102 skills. This dataset is the most dense dataset among all available datasets, because its question–concept density is 1.00% and question–student density is 5.58%.
- **Aaai23**⁴: This dataset was gathered from third-grade math students at Xueersi, a part of the TAL Education Group. It specifically includes data from 18,066 students, 7,652 questions, 1,175 Knowledge Components (KCs), and 5,549,635 interactions or responses. And the question–concept density within this dataset is approximately 0.27% and question–student density of it is about 3.84%.

4.1.2. Baseline

To show the advantages of our STHKT, we contrast our approach with **eight** other cutting-edge methods.

- **BKT** (Yudelson et al., 2013): This method is based on the hidden Markov process, which is a hidden Markov model (HMM) with latent variables. That is, an HMM model (Eddy, 1996) is built for each concept separately, to predict the students' mastery of the specific concept. It is important to note that this model does not take into account the correlation between different knowledge concepts.
- **DKT** (Piech et al., 2015): This method represents the initial application of Deep Neural Networks (DNNs) to Knowledge Tracking (KT), utilizing a Recurrent Neural Network (RNN) model to estimate student performance. In our study, we utilize Gated Recurrent Units (GRUs) (Cho et al., 2014) in place of RNNs for the estimation process.
- **DKVMN** (Zhang et al., 2017): The method uses dynamic key–value memory networks to capture students' knowledge states. And it can dynamically update memory and predict students' future performance in future learning tasks based on their responses and historical information.
- **GKT** (Nakagawa et al., 2019): The method involves treating the concept structure as a graph, and subsequently defining the knowledge tracing task as a node classification problem.
- **AKT** (Ghosh et al., 2020): The monotonic attention mechanism is used in this method to capture the connections within the student's interactions, both current and historical. It is founded on the principle of attention.
- **HawkesKT** (Wang et al., 2021): HawkesKT is a knowledge tracing model based on the Hawkes process, utilizing the self-exciting nature of the Hawkes process to depict the learning process of students.
- **LPKT** (Shen et al., 2021): This model, predicated on logistic regression, forecasts students' mastery of different knowledge concepts. The model calculates the mastery probability of each concept by analyzing students' behavioral data, to help evaluate students' learning progress.

² http://base.ustc.edu.cn/data/ASSISTment/2009_skill_builder_data_corrected.zip

³ http://base.ustc.edu.cn/data/ASSISTment/anonymized_full_release_competition_dataset.zip

⁴ <https://ai4ed.cc/competitions/aaai2023competition>

Table 1

The total num of questions, concepts and students in the real world Dataset.

Datasets	Assistment2009	Assistment2017	Aaai23
Question	17,751	4,117	7,652
Concept	123	102	1,175
Student	4,163	1,709	18,066

- **LBKT** (Xu et al., 2023): The model is grounded in the behavioral data of students, predicting their mastery of new knowledge by simulating the transitions in the knowledge state within the learning process. It is conducive to teachers and students in the education field understand and evaluate students' learning progress, and provide guidance for personalized teaching.

4.1.3. Evaluation metric

AUC (Area Under Curve) is a metric that evaluate the performance of a binary classification model. AUC represents the probability that a randomly chosen positive example has a higher score than a randomly chosen negative one.

ACC (Accuracy) is a metric that measures the proportion of correct predictions among the total number of predictions made by the model. ACC represents how well the model can classify the examples correctly.

4.2. Comparison with state-of-the-arts

In the subsequent experiments, we validate the effectiveness of our model design by comparing the results of different KT models completing the same task. We implemented all the aforementioned baseline methods and our proposed models using PyTorch. All methods were run with the parameters suitably. All models were trained and predicted on an RTX3090 GPU to ensure a fair performance comparison in the interest of fairness.

In the three aforementioned real-world datasets, the task for all methods is to predict the correctness probability of the next question. The prediction accuracy is measured by two common metrics: AUC and ACC. The AUC and ACC test results of various methods on various datasets are presented in Table 3. This table includes the mean and standard deviation of the corresponding indicators. Firstly, the STHKT method performs best on three real-world datasets, indicating its efficacy in modeling both effects and integrating them with time information, thereby enhancing predictions for future questions. Secondly, compared to Assist2009 dataset which is the most sparse, the denser Assist2017 and Aaai23 datasets show more noticeable improvement in the task experiments. This is due to the richer structural information contained within these datasets, thereby amplifying the impact of the two major effects. Finally, compared to the ACC metric results, the improvement in the AUC metric of STHKT is more noticeable. This is because STHKT takes into account the information of the two major effects and handles imbalanced data better.

4.3. Ablation studies

In the section, we have **three** ablation experiments in order to prove the effectiveness of Graph Convolution in the Topological Hawkes Process, two effects and spatiotemporal attention mechanism respectively.

In our experiment, there exists three fixed parameters: The number of batches for a single training iteration, the uniform dimension size used for embeddings throughout the model training process, and the length of the sequence processed by the model at one time are denoted as *batch*, *dmodel*, and *seqlen*, respectively.

For consistency, all ablation methods were run with the parameters *batch*=50, *dmodel*=128, and *seqlen*=200. All models were trained and predicted on an RTX3090 GPU to ensure a fair performance comparison.

4.3.1. Effectiveness of graph convolution in the Hawkes process

This section verifies the effectiveness of integrating graph convolutional networks with Hawkes processes in this model. It accomplishes this by implementing the elimination graph convolutional network within the Hawkes process.

In the method introduction section, it is evident that graph convolution, as proposed by Zhang et al. (2019), is exceptionally well-suited for the integration of structural information with Hawkes process. In this section, we employ a random walk method (Keikha et al., 2018) to replace the graph convolution process. Specifically, we utilize the random walk method, as outlined in Brin and Page (1998), to identify the closest chainlen nodes (where chainlen represents the length of the chain, and is equivalent to the number of random walk steps). Subsequently, we perform a weighted information fusion on the question nodes, taking into account the sequence distance, and ultimately, we arrive at a representation of the question that incorporates pertinent structural information.

The discrepancy between STHKT-nc and STHKT, as illustrated in Table 4, indicates that utilizing the Hawkes process alone to integrate time and space information can enhance the prediction level to a certain extent. However, its capacity to blend structural information and its compatibility with the Hawkes process still lags behind the graph convolution method. Moreover, it still falls short of the predictive capabilities of the combined graph convolution method and the Hawkes process.

4.3.2. Effectiveness of the two effects

This section verifies the role of these two effects in this method by individually and completely eliminating them.

To verify the effectiveness of the two effects, we conducted experiments using the same model structure. We separately employed only the clustering effect, only the group effect, and no effect at all. The results are shown as STHKT-qkc, STHKT-qs, and STHKT-nn in Table 4.

Through these experiments, we found that the STHKT model performance with either the clustering effect or the group effect was superior to that without any effect, but inferior to the model with both effects. This suggests that both the clustering and group effects are beneficial for the task. Moreover, the results showed that the group effect generally outperformed the clustering effect. According to Table 2, this is related to the information density associated with each effect.

4.3.3. Effectiveness of spatiotemporal attention mechanism

This section verifies the important role of spatial information in attention by eliminating spatial information from the spatiotemporal attention mechanism.

In our proposed methodology, we effectively combine temporal and spatial information by utilizing the Hawkes process. This process enhances the attention calculation of the knowledge tracing model, which is based on the opportunistic attention mechanism. To verify the effectiveness of the Hawkes process, we employ a common reinforcement process combination method:

$$\hat{\alpha}_{i,j} = \alpha_{i,j} + W_{qkc} h_{qkc} + W_{qs} h_{qs}, \quad (17)$$

where $\alpha_{i,j}$ is the attention score that solely focuses on the temporal information; h_{qkc} and h_{qs} are the question embeddings that obtain structural information after convolution. These embeddings are simply combined by the trainable matrices W_{qkc} and W_{qs} in terms of time and space.

The distinction between STHKT-nh and STHKT illustrates the remarkable capacity of the Hawkes process to integrate time and space. Utilizing the Hawkes process to amalgamate the two key effects can enhance the accuracy of knowledge tracing predictions.

Table 2
The num of the edges in the real world Dataset.

Datasets	Assistment2009		Assistment2017		Aaai23	
	num	density	num	density	num	density
q-kc	21,246	0.97%	4,192	1.00%	24,867	0.27%
q-stu	278,876	0.38%	392,619	5.58%	5,549,635	3.84%

Table 3

The result of the experiment compared with other advanced knowledge tracing methods. ‘-’ indicates that the attributes present in the dataset do not satisfy the model.

	Assistment2009		Assistment2017		Aaai23	
	AUC	ACC	AUC	ACC	AUC	ACC
BKT	0.6914 ± 0.0144	0.6420 ± 0.0215	0.5523 ± 0.0169	0.6085 ± 0.0154	0.5568 ± 0.0176	0.5031 ± 0.0197
DKVMN	0.7959 ± 0.0007	0.7494 ± 0.0009	0.6913 ± 0.0004	0.6716 ± 0.0120	0.6985 ± 0.0020	0.7901 ± 0.0270
DKT	0.7889 ± 0.0069	0.7586 ± 0.0036	0.6149 ± 0.0027	0.6486 ± 0.0009	0.6055 ± 0.0031	0.7522 ± 0.0008
GKT	0.7806 ± 0.0007	0.7505 ± 0.0024	0.5615 ± 0.0030	0.6165 ± 0.0026	0.6742 ± 0.0025	0.7483 ± 0.0031
AKT	0.8350 ± 0.0007	0.7874 ± 0.0014	0.7415 ± 0.0016	0.7006 ± 0.0004	0.7432 ± 0.0009	0.7932 ± 0.0016
HawkesKT	0.7962 ± 0.0014	0.7572 ± 0.0008	0.7191 ± 0.0002	0.6830 ± 0.0004	0.6102 ± 0.0027	0.7024 ± 0.0008
LPKT	0.7788 ± 0.0032	0.7325 ± 0.0035	0.7360 ± 0.0023	0.6558 ± 0.0014	–	–
LBKT	0.8304 ± 0.0011	0.7860 ± 0.0019	0.6390 ± 0.0050	0.6850 ± 0.0120	–	–
STHKT	0.8391 ± 0.0008	0.7935 ± 0.0025	0.7503 ± 0.0015	0.7073 ± 0.0012	0.7611 ± 0.0016	0.8006 ± 0.0021

Table 4

The ablation experiment.-nh represents the removal of the Hawkes module in the STHKT model, replaced by a regular neural network; -nc indicates the removal of the graph convolution module in the STHKT model, replaced by a random walk with a chain length=9; -nn,-qkc and -qs refer to the cases where both the Hawkes process and the graph convolution module are used, but only the clustering effect or only the population effect is modeled, respectively.

	Assistment2009		Assistment2017		Aaai23	
	AUC	ACC	AUC	ACC	AUC	ACC
STHKT	0.8391 ± 0.0008	0.7935 ± 0.0025	0.7503 ± 0.0015	0.7073 ± 0.0012	0.7611 ± 0.0016	0.8006 ± 0.0021
STHKT-nh	0.8301 ± 0.0017	0.7903 ± 0.0037	0.7427 ± 0.0016	0.7024 ± 0.0007	0.7449 ± 0.0017	0.7943 ± 0.0010
STHKT-nc	0.8239 ± 0.0032	0.7792 ± 0.0033	0.7444 ± 0.0013	0.7034 ± 0.0012	0.7512 ± 0.0020	0.7963 ± 0.0024
STHKT-nn	0.8076 ± 0.0011	0.7659 ± 0.0017	0.7401 ± 0.0002	0.7020 ± 0.0001	0.7248 ± 0.0025	0.7867 ± 0.0022
STHKT-qs	0.8302 ± 0.0024	0.7789 ± 0.0031	0.7497 ± 0.0009	0.6981 ± 0.0012	0.7572 ± 0.0027	0.7993 ± 0.0030
STHKT-qkc	0.8318 ± 0.0004	0.7807 ± 0.0009	0.7443 ± 0.0003	0.7022 ± 0.0016	0.7463 ± 0.0017	0.7953 ± 0.0021

4.4. Case study

4.4.1. Verification of solving the clustering effect of concepts and questions

This section visualizes the embedding correlation of questions for the same student at different time states and analyzes the role of clustering effects.

The STHKT method can cluster questions during the knowledge tracing process. It not only enhances the analysis of student knowledge abilities, but also facilitates the examination of complex relationships between concepts. Concurrently, it can also aid in effective question recommendation.

In this section, we utilize the T-SNE method (Lyu et al., 2019; Van der Maaten & Hinton, 2008), to visualize concept representation vectors. Specifically, we first generate the influence feature vectors $J_{ij} = \frac{y(j|i)}{\sum_i y(j|i)}$, where $y(j|i)$ is the average correct probability assigned by STHKT to problem j when question i is correctly answered (Tong et al., 2020). We then project these vectors into a two-dimensional space and subsequently examine the clustering status of the question–concept relationship at various points in time.

As illustrated in Fig. 4, the progression of question-solving sequences gradually leads to the merging of relationships between questions, showcasing the phenomenon referred to as the clustering effect. Leveraging these clustering outcomes enables a more effective exploration of the causal relationships between concepts, as well as the complex interconnections among questions within the educational domain. This amplifies the advantages of question recommendation to some extent.

4.4.2. Verification of solving the group effect of students and questions

This section analyzes the role of group effects by comparing the visual results of different models on the similarity of results processed by three students with different degrees of association in the same sequence.

To examine the capability of our method in capturing the group effect in knowledge tracing, we employ various models to forecast the performance on a common set of questions for three distinct students: stu_1 , stu_2 , and stu_3 . The test questions are denoted by $T = \{T_1, T_2, \dots, T_{10}\}$. The overlap in their question-solving is calculated using the formula $J(stu_i, stu_j) = \frac{\text{intersection}(stu_i, stu_j)}{\text{union}(stu_i, stu_j)}$. The similarity between stu_2 and stu_1 is 46.55%, while stu_3 and stu_1 have no common questions. As illustrated in Fig. 5, stu_1 serves as the control group, and the predictions that are closer to those of stu_1 are marked. It is evident that, compared to other models, our model generates more similar predictions when evaluating the performance of students with higher similarity. In other words, our method effectively accounts for the question-solving scenario and proficiency levels of other students in the group, reflecting the influence of the group effect among students.

4.4.3. Analysis of improvement effects and sparsity

In this section, we compare the STHKT model with its variants and the best baseline method, assessing the Impact of Sparsity on the Improvement Effects of STHKT.

As shown in Fig. 6, while the improvement effect of STHKT is poor when the dataset density is too low, it increases as the dataset density rises. However, with further increases in dataset density, the improvement effect starts to decrease. We attribute this to the introduction of topological subgraphs, which enhances the edge density of the subgraphs. Additionally, the incorporation of topological graphs means that the analysis of the graph does not rely solely on structure, which reduces STHKT’s dependence on dataset density and mitigates the impact of data sparsity.

Furthermore, we also observe that modeling the two major effects separately results in less improvement compared to when both effects are modeled simultaneously. This indicates a synergistic interaction between the two effects, suggesting that they complement each

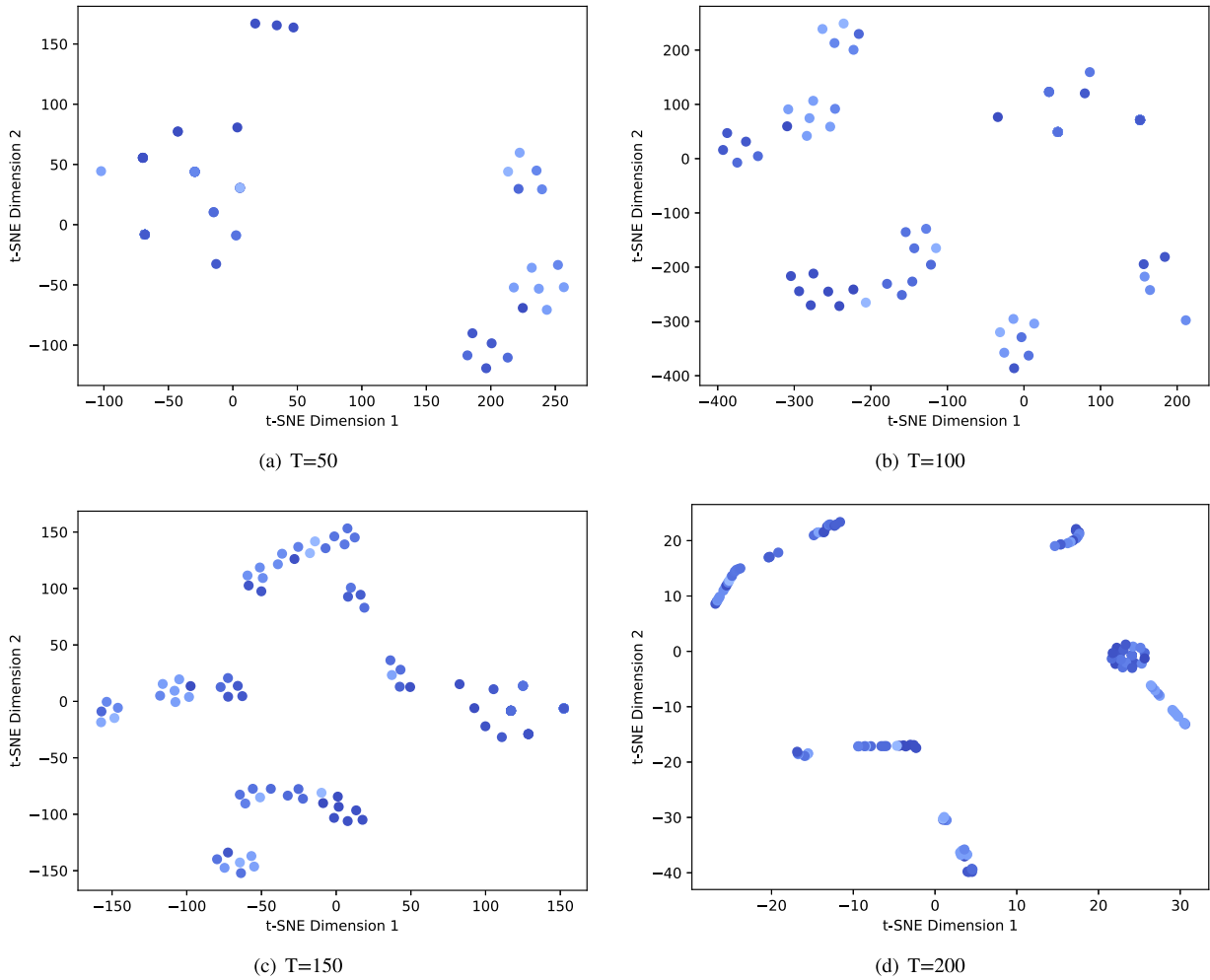


Fig. 4. The trend of question embedding changes after doing a different number of questions. A sequence consisting of 200 interaction records is selected from the Aaa23 dataset. The influence score from question i to question j is calculated through $J_{ij} = \frac{y(i|j)}{\sum_i y(i|j)}$, and then visualized after dimension reduction via the t-SNE method. The visualization is performed at the moments when 50, 100, 150, and 200 questions have been completed in the sequence, respectively. It is observed that as the number of completed questions increases, there appears to be a trend towards clustering of questions.

other and interact in a way that significantly enhances the overall performance and accuracy of the STHKT model. The combined modeling approach thus provides a more comprehensive understanding and better predictive power.

4.4.4. The visualization proof of spatiotemporal attention mechanism

This section visualizes the effects of ordinary attention mechanisms and spatiotemporal attention mechanisms when processing the same sequence, and analyzes the effectiveness of spatiotemporal attention mechanisms.

We conducted a visualization experiment using the Aaa23 dataset, selecting a sequence of five questions, including four different questions.

As illustrated in Fig. 7, our model incorporates a spatiotemporal attention component that utilizes both the standard monotonic attention mechanism and the spatiotemporal attention mechanism. The latter integrates the Hawkes process with two key effects for learning and prediction. After visualizing the attention scores using a heatmap, we observed that the spatiotemporal attention mechanism, which integrates clustering and group effects, no longer excessively focuses on the time-distance relationship of the historical sequence, unlike the monotonic attention mechanism. Unlike the monotonic attention mechanism, it incorporates structural information, resulting in a more diversified calculation of attention scores.

As illustrated in Figs. 7(a) and 7(b), Fig. 7(a) represents the heatmap of attention scores during prediction using only the monotonic attention mechanism in the second stage. On the other hand, Fig. 7(b) represents the heatmap of attention scores during prediction using the spatiotemporal attention mechanism, which combines clustering effects and group effects using the topological Hawkes process in the spatiotemporal module. It is evident that when the model is predicated solely using the monotonic attention mechanism, it primarily focuses on the time series relationship of the question sequence, although it does pay attention to global information. In Fig. 7(a), the attention at a certain moment often pays more attention to the information from the previous moment. However, the spatiotemporal attention mechanism, which combines clustering effects and group effects using the topological Hawkes process, pays attention to both temporal information and the structural information behind the two major effects. In Fig. 7(b), it is evident that the attention at a certain moment also pays more attention to questions that are more structurally related to previous moments. From this, we can infer the optimization of the spatiotemporal attention scores for the model.

5. Conclusion and discussion

In this paper, we integrated two types of dynamic structural information: clustering effect and group effect, into the temporal information of knowledge tracing, using topological Hawkes processes and

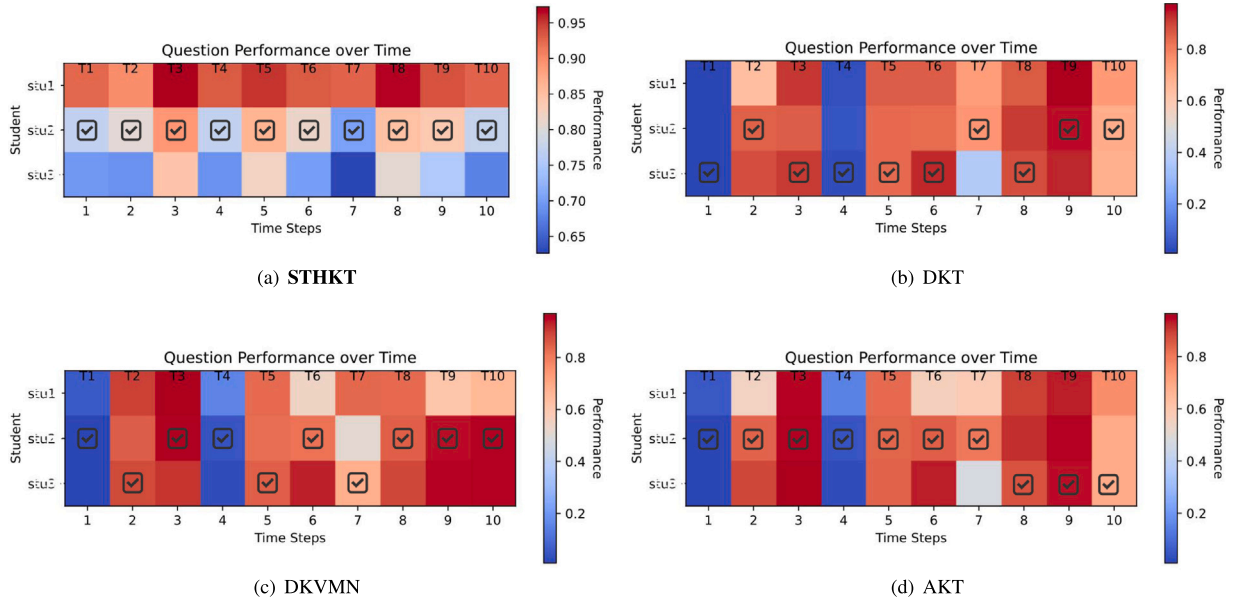


Fig. 5. Four models are employed to predict the outcomes of three students' interaction sequences. Among these, stu_1 serves as the control groups, while stu_2 and stu_3 represent the most similar group and the most dissimilar group, respectively. The similarity between stu_2 and stu_1 is 46.55%, while stu_3 and stu_1 do not share any common questions. Through forecasting the accuracy of specified question sequences using different models, the prediction results of stu_2 and stu_3 that are closer to those of stu_1 will be ticked.

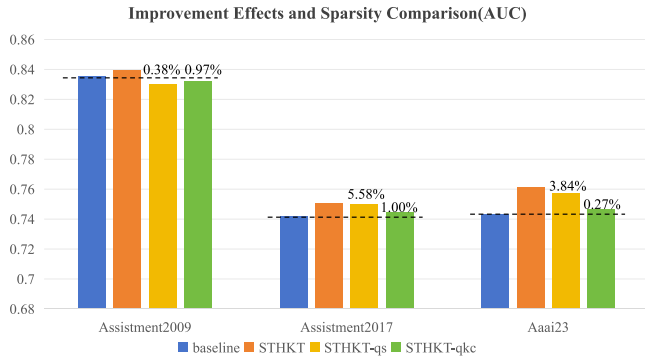


Fig. 6. Comparison of Improvement Effects and Sparsity: We compare the STHKT model, incorporating both effects, with models focusing on clustering (STHKT-qkc) or group effects (STHKT-qs), and the top baseline method. Dashed lines indicate the top baseline results for each dataset. Percentages for STHKT-qkc and STHKT-qs show question-concept and question-student densities respectively.

graph neural networks. This results in the creation of a spatiotemporal knowledge tracing model, STHKT. Firstly, we constructed a tripartite heterogeneous graph structure, establishing relationships among concepts, problems, and students. Subsequently, we topologized this structure to preliminarily integrate temporal and spatial information. Following this, we processed this information within the graph using graph convolution to model the two effects. Finally, we constructed the topological Hawkes intensity function formula, which combined time and dynamic structural information to enhance the monotonic attention mechanism to spatiotemporal attention mechanism. We evaluated the model through a large number of experiments, including comparisons with more advanced knowledge tracing models and ablation experiments on the two most important parts of our proposed model. At the same time, we had **four** case studies to visualize the results. The experiments mentioned before ultimately proved the effectiveness of our model.

Our proposed method has practical implications and applications for the education field as well. It enhances prediction accuracy of students' learning states by using two major effects and a spatiotemporal attention mechanism. It also enables flexible teaching adjustments and

personalized question design by modeling concept and learning trend relationships. Additionally, it integrates well with existing intelligent tutoring systems without needing major infrastructure changes, offering promising research prospects.

However, this research still has some limitations. First, due to the combination of graph convolution and Hawkes processes, this method may lead to higher computational complexity. Although we have argued that this method is computationally feasible, it still has an impact on very large-scale datasets. Additionally, when the dataset is too sparse, modeling the two effects can be less effective. Therefore, we will focus on addressing data sparsity in future researches. We plan to address data sparsity through synthetic data generation, model integration, transfer learning, and large language models. These methods will improve STHKT performance on sparse datasets.

CRedit authorship contribution statement

Shuting Li: Conceptualization, Methodology, Writing – original draft, Review & editing. **Shuanghong Shen:** Conceptualization, Review & editing. **Yu Su:** Methodology, Project administration, Supervision. **Xinjie Sun:** Methodology, Review & editing. **Junyu Lu:** Investigation. **Qi Mo:** Resources. **Zhenyi Wu:** Software. **Qi Liu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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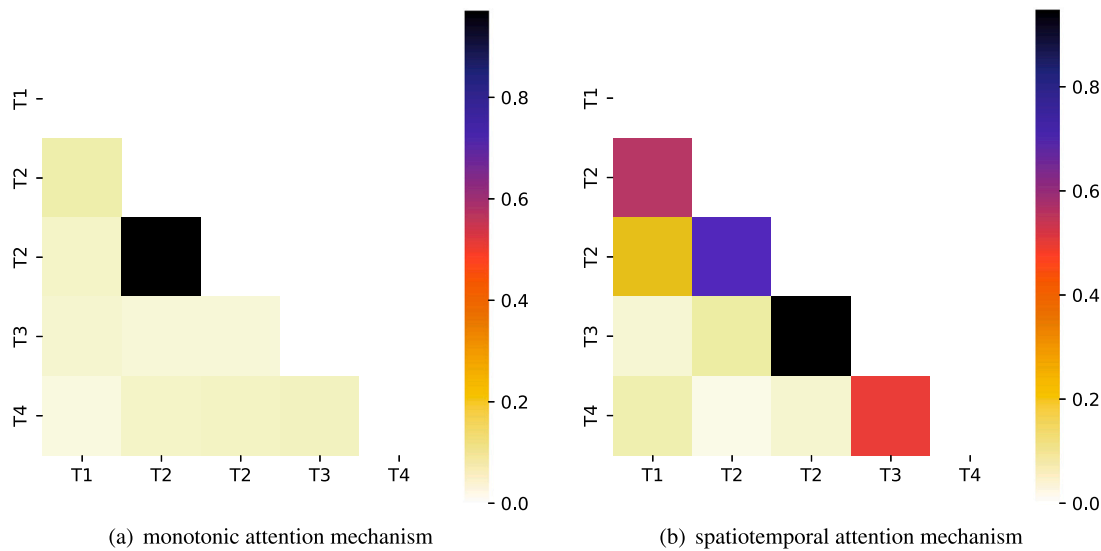


Fig. 7. The visualization of attention scores using different attention mechanisms in the spatiotemporal attention part is presented as a heatmap in the figure. Both the x -axis and y -axis represent the sequence of answering questions, with the direction from left to right and from top to bottom indicating the chronological order. Since the information from subsequent and current moments is not to be referred to at a certain moment, the attention scores in the main diagonal and upper triangular area are all zero. The color bar on the right indicates the variation in color corresponding to the attention scores.

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